

LESSON 5-1 • HOMEWORK PACKET • SHEET 1 OF 2**Area of Parallelograms**

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a parallelogram by multiplying the base by the perpendicular height.

REMEMBER

- Cut a triangle off a parallelogram and slide it over to make a rectangle: $A = b \times h$.
- The height h is **perpendicular** to the base. Do not use the slanted side.
- Area is in **square units** (cm^2 , in^2 , ft^2).

WORKED EXAMPLE

Find the area: base 15 in, slanted side 8 in, height 6 in.

Step 1 Use the perpendicular height: $h = 6$ (skip the 8 in side)

Step 2 $A = b \times h = 15 \times 6$

Step 3 Multiply: $15 \times 6 = 90$

Answer **90 in²**

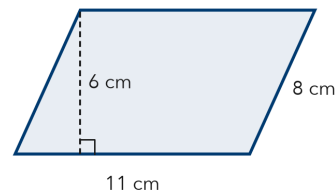
PRACTICE

show your work • label units

1. Find the area of the parallelogram.

Show your work.

Answer _____ cm^2



2. Find the area of each parallelogram with base b and height h .

a. $b = 12 \text{ ft}$, $h = 7 \text{ ft}$ _____ ft^2

b. $b = 9 \text{ m}$, $h = 9 \text{ m}$ _____ m^2

c. $b = 20 \text{ in}$, $h = 4 \text{ in}$ _____ in^2

d. $A = 63 \text{ m}^2$, $h = 7 \text{ m}$; $b =$ _____ m

3. **Parking Lot.** A lot has 10 parking spaces. Each space is a parallelogram with a base of 9 ft and a perpendicular height of 18 ft. What total area do the spaces cover?

Show your work.

Answer _____ ft^2

4. **Mural.** A mural is painted on a parallelogram-shaped wall section. The area is 108 ft^2 and the base is 12 ft. What is the perpendicular height?

Answer _____ ft

5. **Find the Error.** A parallelogram has a base of 10 cm, a slanted side of 7 cm, and a height of 5 cm. Kiara says its area is 70 cm^2 . Explain her mistake and find the correct area.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Find a parallelogram shape at home or on the street (a tile, a sign, a parking space). Show someone where its base and height are.

LESSON 5-1 • HOMEWORK PACKET • SHEET 2 OF 2**Area of Rhombuses**

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a rhombus using the base and height or the two diagonals.

REMEMBER

- A rhombus is a parallelogram, so $A = b \times h$ works.
- With the **diagonals**: $A = (d_1 \times d_2) \div 2$. Do not forget to divide by 2.

WORKED EXAMPLE

Find the area: diagonals 12 cm and 9 cm.

Step 1 Multiply the diagonals: $12 \times 9 = 108$

Step 2 Divide by 2: $108 \div 2 = 54$

Answer 54 cm^2

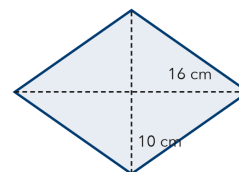
PRACTICE

show your work • label units

1. Find the area of the rhombus.

Show your work.

Answer _____ cm^2

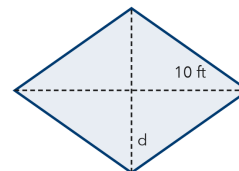


2. Find the area of each rhombus. Choose the formula that fits.

a. diagonals 14 m and 10 m _____ m^2 **b.** base 9 ft, height 7 ft _____ ft^2

3. **Window.** A rhombus window has an area of 60 ft^2 . One diagonal is 10 ft. Find the other diagonal d .

Answer _____ ft



4. **Stretch.** Rhombus A has diagonals of 12 ft and 8 ft. Rhombus B has a base of 8 ft and the same area as Rhombus A. What is the height of Rhombus B?

Answer _____ ft

5. **Find the Error.** Ravi says a rhombus with diagonals of 18 in and 8 in has an area of 144 in^2 . Explain his mistake and find the correct area.

TEST PRACTICE

choose the best answer

6. A rhombus has sides of 7 m and a perpendicular height of 5 m. What is its area?

A 12 m^2 **B** 17.5 m^2 **C** 35 m^2 **D** 49 m^2

7. A rhombus has diagonals of 20 in and 9 in. What is its area?

A 29 in^2 **B** 45 in^2 **C** 90 in^2 **D** 180 in^2

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____